

PRATHAM P PANSARE

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ABOUT ME

Aspiring software developer with hands-on experience building scalable, full-stack applications and AI-driven platforms. Proficient in modern web frameworks, relational databases, and machine learning integration, with a strong ability to translate complex technical requirements into user-centric solutions. Passionate about software architecture, clean code, and continuous learning.

EDUCATION

A. P. Shah Institute of Technology (Engineering), Thane

B.Tech in Information Technology

Jana Gana Mana College, Dombivli

Higher Secondary Education (11th - 12th)

Model English School, Dombivli West

Secondary Education (1st - 10th)

SKILLS

Languages: Python, JavaScript, TypeScript, CSS, HTML, SQL

Libraries/Frameworks: React, Next.js, FastAPI, Flask, Tailwind CSS, Bootstrap, Prisma

Databases: MySQL, PostgreSQL, SQLite, Supabase, Redis

Cloud & DevOps: AWS, Cloudflare R2, GitHub Actions, Vercel, Hugging Face Spaces

Tools: TensorFlow, D3.js, Figma, Canva, Google Colab, GitHub, Capacitor

PROJECTS

ShetVaidya — AI Crop Disease Analyzer & Medicine Verifier React, FastAPI, TensorFlow, PostgreSQL

- Built a production-grade, location-aware crop disease analyzer and medicine verification platform (ShetVaidya = "field doctor" in Marathi) to empower farmers with immediate diagnostic tools.
- Developed a TensorFlow MobileNetV2 inference microservice with V2.0 prediction safety — dedicated non-leaf/garbage class rejection and confidence gating for out-of-scope images.
- Engineered a microservice architecture (frontend, backend, ML service) with FastAPI, JWT HttpOnly cookies, Google OAuth, and API rate limiting on auth and batch verification flows.
- Configured Cloudflare R2 object storage for secure image uploads (5MB limit, MIME validation) and integrated ZXing barcode scanning for medicine batch verification with non-public batch codes.
- Established CI/CD via GitHub Actions for containerized deployment across Vercel (frontend) and Hugging Face Spaces (backend + ML), with PostgreSQL and Redis for structured data and caching.

Khatawali Records — Credit/Debit Ledger App

React, Supabase, Capacitor, Bootstrap

- Engineered a mobile-first credit/debit ledger application for small business owners and individuals, with category-separated tracking for the same person across independent ledgers.
- Packaged natively for Android using Capacitor with deep-link OAuth callbacks, in-app contact importer, and biometric fingerprint lock configurable via app settings.
- Designed a PostgreSQL schema on Supabase with Row Level Security (RLS) policies and Supabase Auth (Google OAuth) to guarantee per-user data isolation and privacy.
- Built a dashboard with advanced search (name, phone, description, category), date-range and year/month filtering, and quick actions for call, WhatsApp, receive, give, and transfer.
- Integrated per-person PDF and Excel export using jsPDF, jspdf-autotable, and PapaParse, along with profile/business details, bank holidays, recycle bin, and developer support module.

Scholar Nexus — AI Research Discovery Platform

Next.js, TypeScript, Prisma, z-ai SDK

- Architected an AI-powered research paper discovery platform that simultaneously queries 9 academic databases (Semantic Scholar, arXiv, Crossref, PubMed, OpenAlex, and more) in parallel.
- Implemented a 10-step search pipeline: query understanding, AI-driven expansion, parallel fan-out, normalization, deduplication (DOI + title-hash + fuzzy), ranking, pagination, and AI insights.
- Developed composite ranking scoring papers across citation impact, recency, source authority, and open-access availability, with AI-generated per-paper insights (TL;DR, methodology, limitations).
- Built interactive D3.js citation network graphs, PDF full-text Q&A chatbot, evidence synthesis across papers, color-coded reading collections, daily/weekly search alerts, and author profiles.
- Engineered 18 API route handlers with rate limiting (token bucket), SSRF protection, Zod input validation, and Prisma parameterized queries; deployed on Vercel with GitHub Actions CI.

Air Pollution Tracker V2.0 — Environmental Intelligence

React, Tailwind CSS, Supabase, React Leaflet

- Created a data-driven React application visualizing live and historical air-quality metrics for 200+ cities worldwide, with OpenAQ v3 integration and EPA breakpoint AQI computation.
- Implemented differentiated role-based portals: citizen experience (dashboard, AQI tracking, map view) and government portal (heatmaps, incident desk, pollutant intelligence, reporting panels).
- Built interactive map views with React Leaflet clustered markers and geolocation (browser API with IP-based fallback via ipwho.is), plus Phase 5 analytics for single-city deep dives and forecasting.
- Designed a robust data layer using React Query v5 with stale-while-revalidate caching, 30-minute localStorage cache for OpenAQ snapshots, and virtualized list rendering via react-window.
- Integrated Supabase Auth with domain-guardrailed registration (citizen: @gmail/@outlook, government: @gov), email OTP verification, offline-friendly preferences, and audit logging for government access.